A16727W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: ATOMIC AND LASER PHYSICS

TRINITY TERM 2022

Wednesday, 15th June, 2:30 pm – 4.30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Describe the LS-coupling scheme for angular momentum, giving the conditions for which it is a good approximation. Give the selection rules for electric-dipole transitions in atoms for this case, distinguishing between rules that apply generally and those specific to this angular momentum coupling scheme. What terms and levels arise for the electronic configurations $5s^2$, 5s5p and 5s6s of the strontium atom? [7]

(b) Strong transitions between these levels occur at wavelengths: 461, 679, 688, 707 nm and 1.12 μm. The 461 nm transition can be observed in absorption. A relatively weak transition at 689 nm can be driven by laser excitation from the ground configuration, to one of these levels. Draw a labelled energy level diagram, marking the energy of each level (in units of m^{-1}) together with the transitions and their wavelengths. Describe your reasoning for these assignments and any assumptions that you make. Which of these lines exhibits a Normal Zeeman effect in an applied magnetic field? [8]

(c) What terms and levels arise for the 5s5d configuration? List all the allowed transitions between these levels and those of the $5s^2$, 5s5p and 5s6s configurations. Which one of these transitions splits into 6 components when observed perpendicular to an applied magnetic field? Explain your reasoning. [6]

(d) Comment on the practical considerations in measurements of this Zeeman splitting with a brief description of a suitable experimental apparatus. [4]

2. (a) Show that an interaction of the form $A\,\mathbf{I}\cdot\mathbf{J}$ leads to an interval rule, where $\mathbf{I}$ and $\mathbf{J}$ are angular momentum operators. What are the physical origins of interactions of this form in atoms that determine structures on various energy scales? In each case, specify what set of good quantum numbers characterise the energy levels arising from these interactions. [8]

(b) The level $4f^7 6s6d\ {}^8D_{11/2}$ of europium is found to have a relatively large hyperfine structure. Explain this. [3]

(c) Europium has two stable isotopes with atomic masses 151 and 153. In a sample containing both isotopes, a transition from a level that has negligible hyperfine splitting to the $4f^7 6s6d\ {}^8D_{11/2}$ level is observed to be split into 12 components at relative positions:

Peak	Position (GHz)
a	21.19
b	18.37
c	14.84
d	10.99
e	10.60
f	9.74
g	8.17
h	6.29
i	5.63
j	4.09
k	1.58
l	0.0

Show that the spacings between the frequencies a,b,c and e obey an interval rule. Hence determine which other peaks arise from this isotope (^{151}Eu). Deduce the nuclear spin of this isotope, explaining your reasoning. [9]

(d) What is the nuclear spin of the other isotope (^{153}Eu)? Determine the ratio of the nuclear magnetic dipole moments of the two isotopes. [5]

3. In a simple model of an atom, the Hamiltonian $\hat{H}_0$ has two eigenstates with energies E_1 and E_2, and $E_2 - E_1 = \hbar\omega_0$. Spontaneous emission is neglected. This system is subjected to a time-dependent external field such that

$$\hat{H} = \hat{H}_0(\mathbf{r}) + \hat{D}_0 \cos(\omega t) \ .$$

The matrix elements $\langle\psi_1|\hat{D}_0|\psi_1\rangle$ and $\langle\psi_2|\hat{D}_0|\psi_2\rangle$ are both zero. The general solution of the time-dependent Schrödinger equation for this system can be written as

$$\Psi(\mathbf{r}, t) = c_1(t)\psi_1(\mathbf{r})\mathrm{e}^{-\mathrm{i}E_1 t/\hbar} + c_2(t)\psi_2(\mathbf{r})\mathrm{e}^{-\mathrm{i}E_2 t/\hbar} \ .$$

(a) Consider the case where the oscillating field is switched on at time $t = 0$ with a constant amplitude (so that $\hat{D}_0$ is constant for $t > 0$). The initial conditions are $c_1 = 1$ and $c_2 = 0$. *Without assuming that the radiation is weak*, calculate the probability to be found in the upper state at time t, showing that when $|\omega - \omega_0| \ll \omega_0$ it has the form

$$|c_2(t)|^2 = U \sin^2(\alpha t) \, .$$

Give expressions for U and α in terms angular frequencies that characterise the system. Calculate the time-averaged value of $|c_2(t)|^2$, averaged over a time interval τ, and give its limiting value as $\tau \to \infty$. [9]

(b) For what conditions do Rabi oscillations occur? Compare these with the conditions for which the Einstein treatment gives a good description of atoms interacting with radiation. [4]

(c) Using a Fourier transform or otherwise, show that lifetime (or natural) broadening has a Lorentzian line shape. Give an example of another broadening mechanism that produces a Lorentzian line shape. [5]

(d) Consider now the case where a time-dependent field (with a maximum amplitude at $t = 0$) is applied such that

$$\hat{H} = \hat{H}_0(\mathbf{r}) + \hat{D}_0 e^{-b|t|} \cos(\omega t) \ ,$$

with $c_1 = 1$ and $c_2 = 0$ at $t = -\infty$. ($\hat{D}_0$ is constant; b is positive and real.) *Assuming that the radiation is weak*, find the amplitude $c_2(t)$ for $t > 0$. Find the excitation probability at long times, given by $|c_2(t)|^2$ as $t \to \infty$, and show that it has the form of a Lorentzian function squared. What is its full width at half maximum (FWHM)? [7]

4. A homogeneously broadened transition of angular frequency ω_{21} occurs between two levels which have degeneracies g_1 and g_2. The upper level is populated at a rate R_2 and decays only by spontaneous emission to the lower level at a rate A_{21}. The lower level is populated only by this decay and has a lifetime τ_1. The number densities in the lower and upper levels are N_1 and N_2 respectively. A beam of radiation with intensity I (at angular frequency ω) propagates through this laser medium.

(a) Give the rate equations for the number densities of the two levels in terms of the gain cross section σ. Show that the gain coefficient at line centre has the form

$$\alpha = \frac{\alpha_0}{1 + I/I_\mathrm{s}} \,.$$

Give expressions for α_0 and I_s, and state what they represent. [9]

(b) Show that the input intensity I_in of a laser amplifier of length L and the output intensity I_out are related by

$$\ln\left(\frac{I_\mathrm{out}}{I_\mathrm{in}}\right) + \frac{I_\mathrm{out} - I_\mathrm{in}}{I_\mathrm{s}} = \alpha_0 L.$$

[3]

(c) Two identical laser amplifiers, with the same α_0, I_s and L, are placed so that a beam of light passes through both. This monochromatic laser beam has intensity I_0 at the input to the first amplifier and is amplified to $70\,I_0$ at its output. This light enters the second amplifier and is amplified to $150 I_0$ at the output. Deduce the value of the ratio I_0/I_s. [6]

(d) By adapting the rate equations for a laser system, or otherwise, write down a set of rate equations that describes the absorption of light by a two-level system. Sodium atoms absorb light on the 3s $^2\mathrm{S}_{1/2}$ to 3p $^2\mathrm{P}_{3/2}$ transition, and the excited level has a lifetime $\tau_{3\mathrm{p}} = 16\,\mathrm{ns}$. A laser beam travels a distance of 1 mm through a sample of cold sodium atoms; 90% of a low intensity beam is transmitted and the transmission is 96.7% for a much stronger beam of intensity $I = 270\,\mathrm{W\,m^{-2}}$. In both cases the frequency is at the centre of the spectral line at 589 nm. Calculate the absorption cross-section. Comment on its size relative to the cross-section for collisions between atoms that cause collisional broadening of spectral lines. What are the conditions for which this transition is homogeneously broadened? [7]